

GRATIOT, MI, USA

WMO: 725405

Lat: 43.322N

Lon: 84.688W

Elev: 747

StdP: 14.30

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54816

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.1	(c) 2.8	(d) -8.4	(e) 3.6	(f) -1.7	(g) -3.1	(h) 4.8	(i) 5.3	(j) 22.9	(k) 22.2	(l) 20.3	(m) 24.3	(n) 2.9	(o) 240	(p) 0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	89.2	72.3	86.2	71.4	83.4	69.5	76.2	84.0	74.3	81.8	72.6	79.8	9.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	127.7	80.1	72.2	122.6	78.7	70.3	114.7	76.6	40.2	83.3	38.2	81.4	36.7	79.3	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.7	18.4	16.5		-10.6	92.3	5.9	2.8	-14.8	94.4	-18.3	96.1	-21.6	97.7	-25.8	99.7	
				DB												
				WB	-11.0	79.0	5.7	2.3	-15.1	80.7	-18.4	82.0	-21.6	83.4	-25.7	85.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	22.5	23.6	33.8	45.9	57.5	66.3	70.3	68.2	61.3	49.8	38.7	28.0
	DBStd	19.02	10.91	10.64	11.12	9.04	8.44	6.84	6.00	5.41	7.52	8.56	8.78	9.57
	HDD50	3468	852	739	517	183	23	0	0	0	6	113	353	682
	HDD65	6968	1316	1158	969	575	264	65	20	29	159	477	789	1147
	CDD50	2475	0	0	14	60	255	489	628	563	344	107	14	1
	CDD65	501	0	0	1	2	31	104	183	127	47	6	0	0
	CDH74	5050	0	0	8	46	344	1045	1871	1177	500	59	0	0
	CDH80	1469	0	0	1	6	76	314	620	313	131	8	0	0
Wind	WSAvg	7.0	7.9	7.9	7.9	8.7	7.4	6.1	5.3	4.9	5.4	6.9	7.6	7.8
Precipitation	PrecAvg	31.1	1.8	1.4	2.1	2.9	3.1	3.2	2.6	3.5	3.4	2.6	2.7	2.0
	PrecMax	42.0	3.9	4.4	6.1	7.3	7.9	6.8	7.0	10.0	16.3	6.0	6.3	4.9
	PrecMin	22.3	0.2	0.0	0.2	0.9	0.3	0.7	0.4	1.1	0.0	0.2	0.5	0.3
	PrecStd	4.8	1.0	0.9	1.1	1.2	1.7	1.5	1.4	1.7	2.6	1.3	1.5	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	56.3	72.5	80.5	86.2	90.5	92.7	90.7	89.3	80.8	68.3	56.9
		MCWB	50.4	50.7	61.2	61.5	69.5	72.7	74.2	72.9	68.8	68.8	56.9	53.7
	2%	DB	45.2	46.4	63.1	72.9	81.8	87.7	89.8	86.7	83.7	74.6	61.3	48.4
		MCWB	42.2	42.7	54.4	57.6	65.9	71.4	73.8	72.1	68.8	62.9	53.4	46.1
	5%	DB	39.4	41.4	55.4	68.0	78.2	83.7	86.4	83.3	80.0	69.9	56.7	44.8
		MCWB	36.9	37.7	49.1	55.2	63.8	69.1	72.3	70.2	67.0	61.1	50.8	42.4
	10%	DB	36.3	37.0	50.3	63.0	73.2	80.7	83.2	81.0	75.5	64.8	52.7	39.5
		MCWB	34.6	34.3	44.6	52.2	61.4	67.6	70.0	68.5	65.2	58.0	47.9	36.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	51.6	62.4	66.0	72.5	76.5	79.6	77.8	75.4	70.8	59.8	54.7
		MCDB	51.9	53.7	71.1	76.3	81.7	85.5	88.2	86.6	82.3	78.1	64.9	57.0
	2%	WB	42.9	43.8	56.0	60.9	69.5	74.0	77.0	75.2	72.6	66.8	55.4	46.7
		MCDB	45.0	47.5	61.5	67.9	76.8	82.6	85.0	82.0	78.6	71.0	60.8	48.1
	5%	WB	37.7	38.3	51.1	57.2	66.8	71.8	74.8	73.1	70.3	62.8	51.9	42.4
		MCDB	39.6	41.0	55.7	65.7	73.9	80.3	82.5	79.6	75.7	67.1	56.6	44.6
	10%	WB	34.7	34.6	44.6	53.3	64.1	69.8	73.0	71.2	67.6	58.9	47.8	37.7
		MCDB	36.2	37.0	48.7	61.6	70.6	77.0	80.3	77.6	73.2	64.4	51.9	39.7
Mean Daily Temperature Range	5% DB	MDBR	13.4	15.5	17.6	20.8	21.3	21.6	22.6	22.0	22.9	19.5	15.2	12.1
		MCDBR	14.8	17.4	25.5	29.1	26.0	24.3	24.6	24.9	27.2	26.4	21.9	16.0
		MCWBR	13.1	13.9	17.2	16.0	13.2	11.4	11.7	12.2	14.1	15.6	15.4	14.2
	5% WB	MCDBR	14.3	16.4	23.1	24.5	21.9	20.8	20.7	20.9	21.0	21.9	20.0	15.8
MCWBR		13.1	13.9	17.3	16.6	13.3	11.5	11.4	12.1	12.8	15.8	16.0	14.6	
Clear-Sky Solar Irradiance	taub		0.289	0.327	0.330	0.370	0.397	0.422	0.411	0.405	0.377	0.338	0.309	0.286
	taud		2.360	2.241	2.382	2.300	2.275	2.238	2.274	2.288	2.359	2.469	2.493	2.424
	Ebn at Noon		267	272	289	284	278	270	272	270	269	268	260	259
	Edh at Noon		25	34	33	39	42	43	41	39	34	27	22	22
All-Sky Solar Radiation	RadAvg		440	722	1142	1450	1702	1915	1939	1650	1324	830	503	352
	RadStd		39	85	122	128	160	114	99	81	125	88	65	35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page